

E-Commerce Sales & Profitability Analysis

Business Performance Report — End-to-end analytics using Python, SQL Server, and Power BI

SUPERSTORE DATASET

9,994 TRANSACTIONS



Made with GAMMA

Executive Summary

Where is the business generating revenue, where is it generating profit, and where are margins being eroded?

\$2.3M

Total Sales

\$286K

Total Profit

5,009

Orders

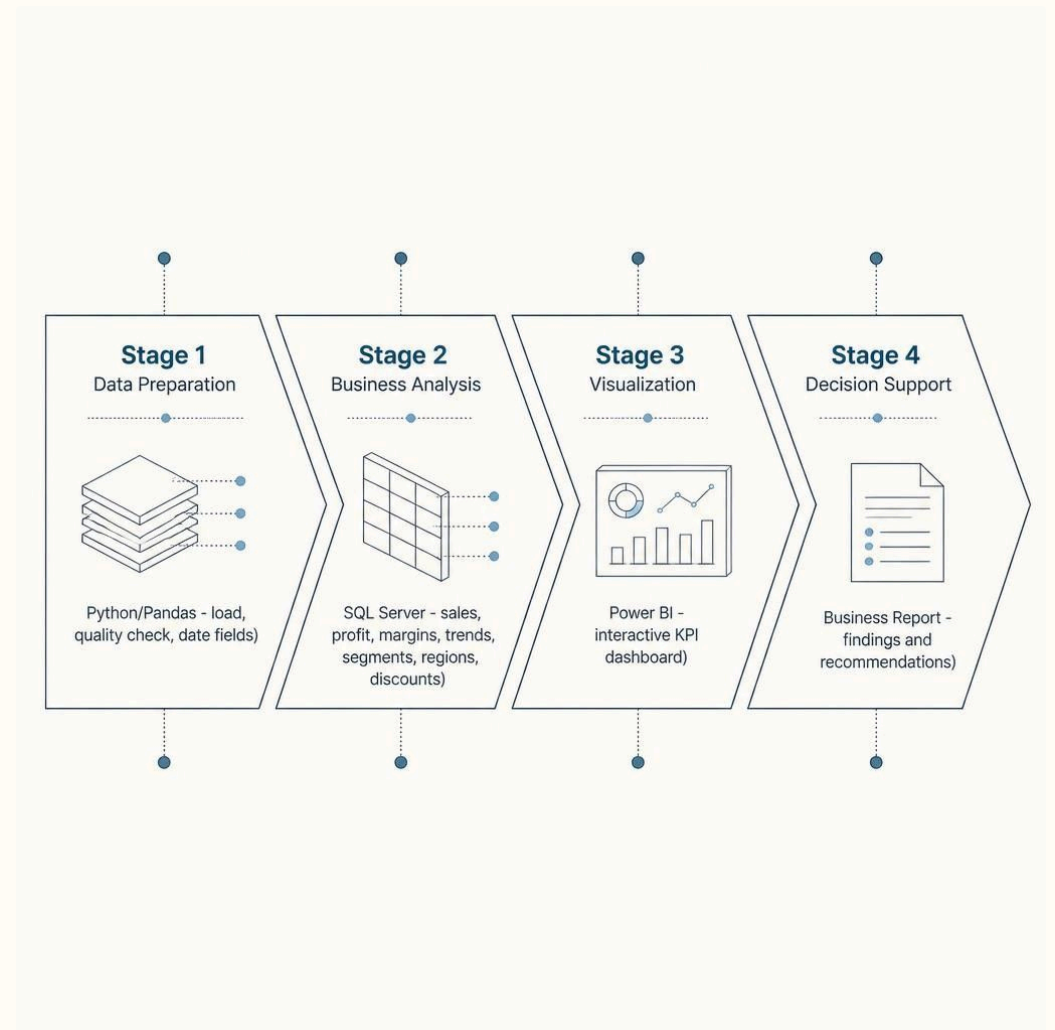
12.5%

Overall Margin

📌 **Key finding:** Sales grew from \$484K (2014) to \$733K (2017), but discounts above 30% produced negative margins. West led regions; Tables, Bookcases, and Supplies were loss-making.

Analytical Approach

A four-stage workflow combining Python, SQL Server, and Power BI — from raw data to management-ready insights.



01

Data Preparation

Python / Pandas — load, quality check, date fields

03

Visualization

Power BI — interactive management dashboard

02

Business Analysis

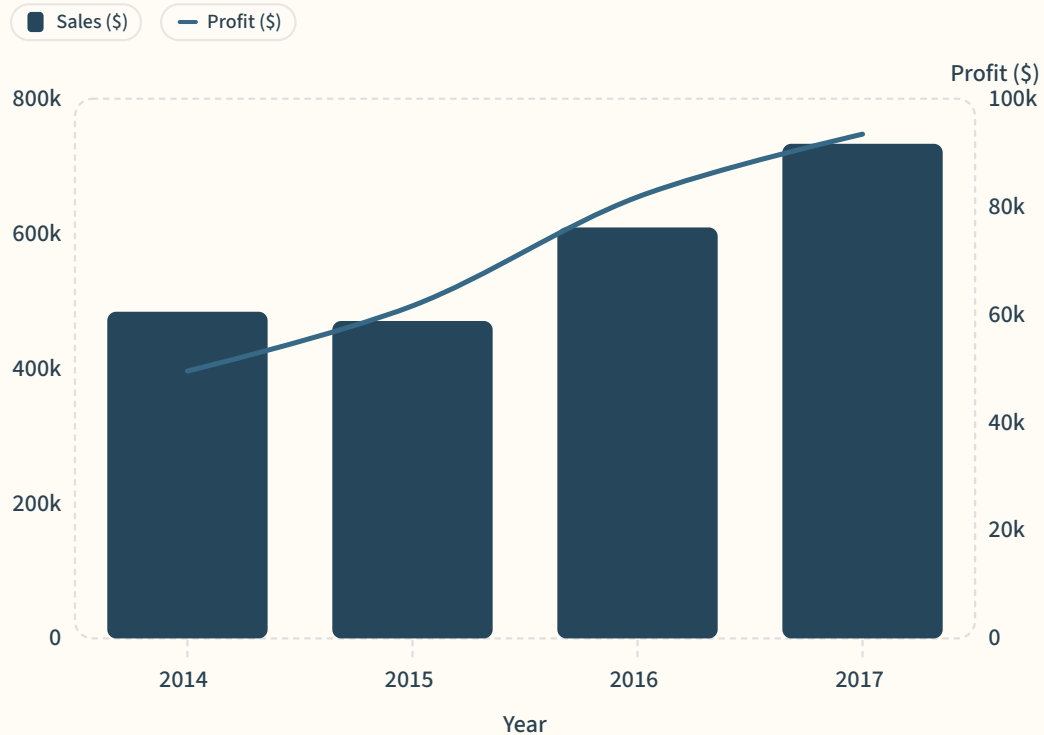
SQL Server — sales, profit, margins, trends

04

Decision Support

Business report — findings and recommendations

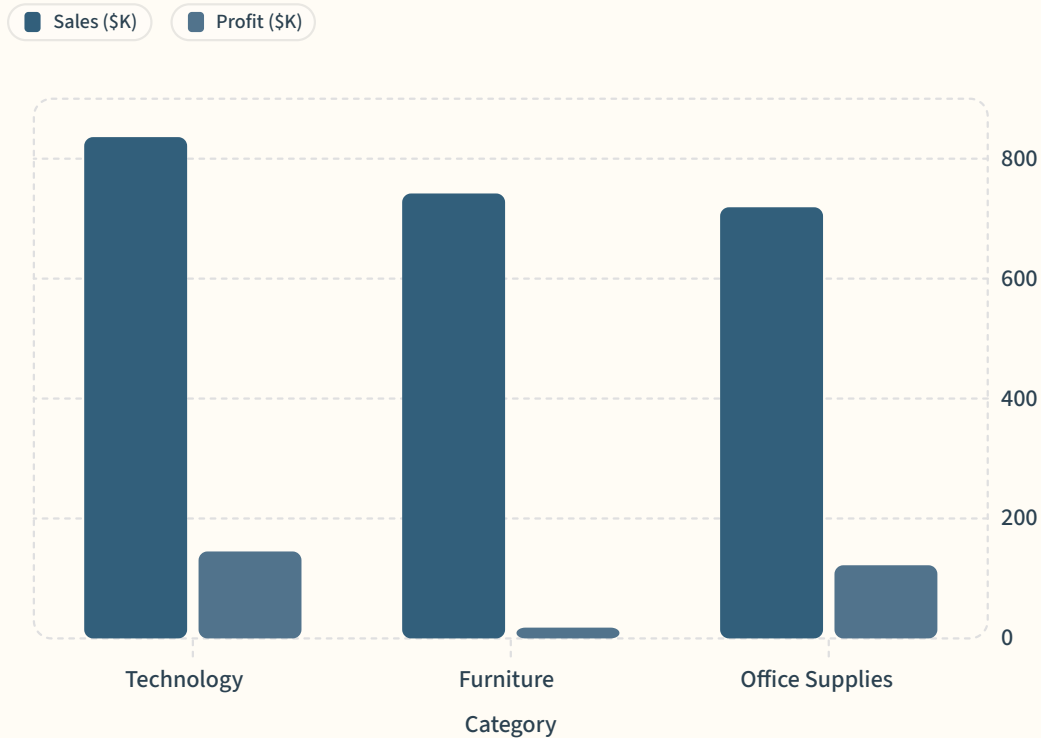
Sales & Profitability: Annual Trend



Key Trends

- Sales grew **51%** from 2014 to 2017, reaching \$733,215
- Largest sales jump: **29.5%** in 2016
- Profit rose from \$49,544 to \$93,439 over the period
- Peak month: **November 2017** (\$118,448 sales)
- Lowest month: **February 2014** (\$4,520 sales)

Category Performance



Category Insights

- **Technology:** \$836K sales, \$145K profit, **17.4% margin**
- **Office Supplies:** \$719K sales, \$122K profit, **17.0% margin**
- **Furniture:** \$742K sales but only \$18K profit — **2.5% margin**

⚠ **Loss-making sub-categories:** Tables (-8.6%), Bookcases (-3.0%), Supplies (-2.5%). Investigate at SKU level for pricing and assortment decisions.

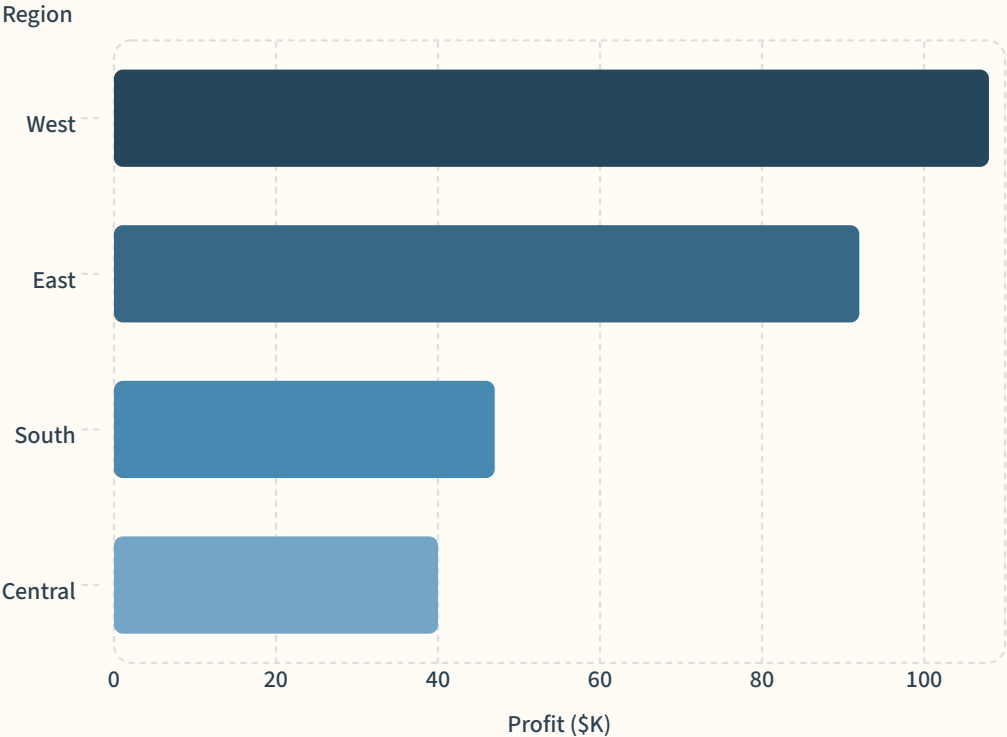
Customer Segments & Regional Performance

Customer Segments

Segment	Sales	Profit	Margin
Consumer	\$1,161,401	\$134,119	11.5%
Corporate	\$706,146	\$91,979	13.0%
Home Office	\$429,653	\$60,299	14.0%

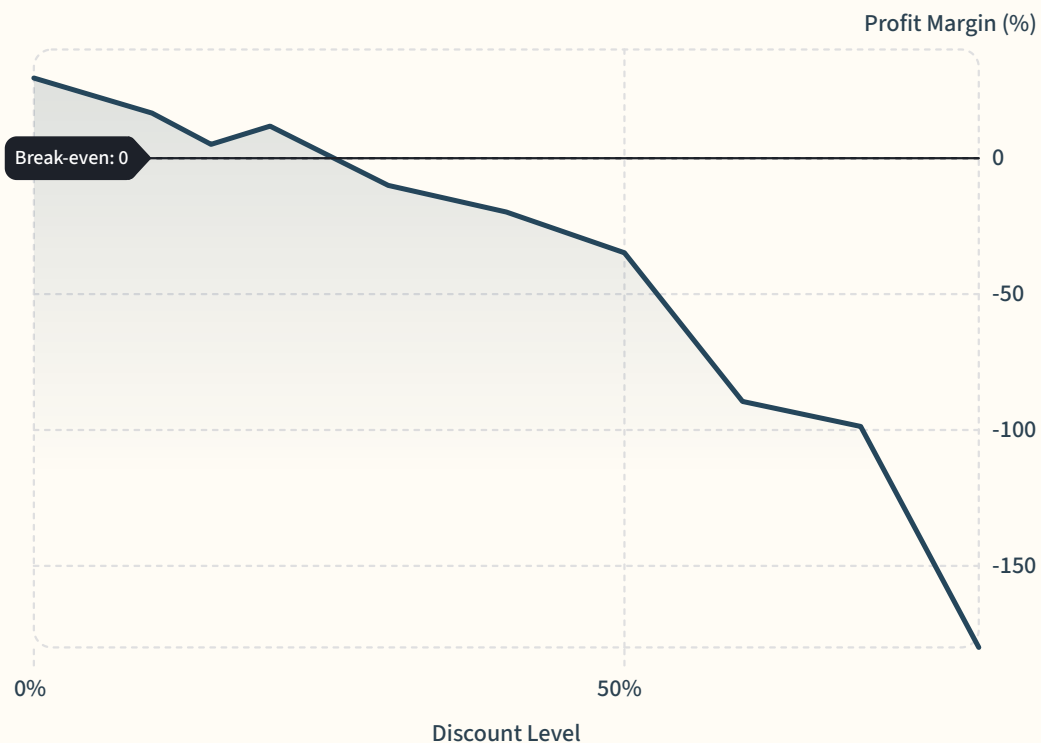
Consumer leads in scale; **Home Office has the highest margin at 14.0%.**

Profit by Region (\$K)



West is the strongest market (14.9% margin). Central's 7.9% margin signals an opportunity to review product mix and discounting.

Discount Impact on Profitability



Key Insight

Profitability **deteriorates sharply** as discounts rise. Aggregate margin turns **negative at 30% discount** and worsens dramatically beyond that point.

- 0% discount → **29.5% margin**
- 20% discount → **11.8% margin**
- 30%+ discount → **negative margins**



Discounts of 30% and above destroyed aggregate margin in this dataset.

Shipping & Top Products

Shipping Mode Performance

Ship Mode	Orders	Sales	Margin
Standard Class	2,994	\$1,358,216	12.1%
Second Class	964	\$459,194	12.5%
First Class	787	\$351,428	13.9%
Same Day	264	\$128,363	12.4%

First Class recorded the highest margin (13.9%). Standard Class dominates by volume.

Top 3 Products by Sales

Canon imageCLASS 2200

\$61,600 sales · \$25,200 profit

Fellowes PB500 Binding

\$27,453 sales · \$7,753 profit

Cisco TelePresence EX90

\$22,638 sales · **-\$1,811 loss**

Management Recommendations

1. Protect Margins

Introduce discount guardrails by category. Require evidence that volume compensates for margin loss.

2. Fix Loss-Makers

Prioritize Tables, Bookcases & Supplies for SKU-level pricing and assortment review.

3. Improve Furniture

Substantial sales but only 2.5% margin — identify products and discount levels driving leakage.

4. Replicate West

Use West (14.9% margin) as a benchmark for product mix and discount behavior in weaker regions.

5. Profitable Growth

Use segment-specific pricing. Home Office shows smaller segments can have stronger margins.

6. Plan Seasonality

Use stronger year-end demand to plan inventory, promotions, and capacity ahead of peak periods.

Conclusion & Deliverables

Strategic Takeaway

The business shows strong growth, but **revenue growth does not automatically translate to healthy profitability**. The priority is **margin quality**: control excessive discounting, address loss-making product groups, and use stronger regions and segments as benchmarks.

📌 **Priority:** Shift management focus from sales growth alone to **profitable sales growth**.

Project Deliverables

→ Python

Data loading, quality checks, date preparation

→ SQL Server

12 business questions across trends, products, regions, discounts

→ Power BI

Interactive dashboard: KPI, trend, category, region views

→ Business Report

Management-focused findings and recommendations